

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Equipping large language models (LLMs) with the capacity to actively retrieve external knowledge is indispensable for the complex, real-world tasks whose solutions lie beyond the reach of any model’s static parametric memory. Yet prevailing approaches remain confined to two unsatisfactory extremes: prompting-based methods merely elicit a model’s latent agentic behavior without ever strengthening it, whereas reinforcement learning (RL) applied to open-ended interactive search remains beset by performance ceilings and catastrophic collapse, leaving the underlying agentic potential largely untapped. We resolve both failure modes with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm engineered for robust agent training through the tightly coordinated integration of sequence-level optimization and dynamic outcome-based sample filtering—two mechanisms that respectively govern the stability and the informativeness of every policy update. Trained end-to-end through RL alone to interleave multi-turn search and reasoning, DSPO dispenses entirely with supervised demonstration data. Across seven question-answering benchmarks, our 7B model improves over a comparable prior method by **34.1%** and, more strikingly, surpasses the 14B model of that same prior work on complex multi-hop QA such as HotpotQA by nearly **9% relative**—all while sustaining exceptional training stability. These results establish that principled algorithm design, rather than sheer model scale, is the decisive lever for unlocking reliable agentic search.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have attained remarkable performance across a broad spectrum of specialized tasks, spanning mathematics (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code generation (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). This proficiency, however, is circumscribed by a fundamental limitation: the knowledge encoded in an LLM is inherently static, bounded by the corpus on which it was trained. The most compelling remedy for this knowledge cutoff is to endow LLMs with search capabilities, transforming them into agents that actively interrogate external knowledge sources (Jin et al., 2025b). Such a capability constitutes a paradigmatic instance of tool use (Schick et al., 2023), in which the model learns to interact with an external search engine so as to resolve questions that parametric memory alone cannot answer. Mastering it demands the acquisition of a complex, multi-step policy—a requirement that casts the problem squarely as sequential decision-making, precisely the regime for which Reinforcement Learning (RL) is designed.

Unlike Supervised Fine-Tuning (SFT), which depends on costly static demonstrations and is constitutionally incapable of teaching exploration, RL furnishes a principled framework in which LLMs acquire effective policies through trial-and-error interaction with their environment (Chu et al., 2025). Value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have consequently emerged as the dominant paradigm, prized for their simplicity and their markedly reduced memory footprint. Yet however successful these methods prove on constrained tasks, transposing them to the open-ended domain of interactive search exposes critical instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). This fragility is not incidental; it traces directly to two fundamental and mutually compounding flaws. First, as established by Zheng et al. (2025), GRPO’s token-level objective is ill-posed when paired with a sequence-level reward, and the resulting mismatch produces high-variance gradient estimates that destabilize training. Second, the sparse rewards intrinsic to search tasks frequently yield sample groups with homogeneous outcomes—uniformly successful or uniformly failed—collapsing the advantage signal to zero, starving the optimizer of any usable gradient, and severely degrading sample efficiency (Yu et al., 2025; Liu et al., 2025). Acting in concert, these two pathologies confine existing algorithms to a regime of unstable or stalled learning; it is precisely this regime that our method is engineered to escape.

To confront these two failure modes directly and simultaneously, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an algorithm that synthesizes and refines complementary principles from recent policy optimization research within a single coherent framework. To eliminate the reward–optimization mismatch, DSPO adopts the sequence-level formulation of GSPO (Zheng et al., 2025), matching the unit of optimization to the unit of reward and thereby aligning the objective with the trajectory-level signal that genuinely governs credit assignment—an alignment that fundamentally stabilizes learning over long-horizon reasoning. To eliminate the vanishing-signal problem, DSPO further incorporates a dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025), which constructs each training batch exclusively from rollout groups containing both successful and unsuccessful trajectories for a given prompt, thereby guaranteeing that the advantage signal $\hat{A}_i$ remains well-defined and informative at every update. Critically, these two components are mutually reinforcing rather than merely additive: sequence-level updates furnish the stability, dynamic filtering supplies the dense signal, and their conjunction yields a stable, high-performance algorithm purpose-built for complex, multi-turn search and reasoning; as our ablations confirm, neither component alone suffices. Trained under this unified framework, our 7B model attains a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b) and, more strikingly, surpasses its 14B counterpart (Jin et al., 2025a) on complex multi-hop benchmarks such as HotpotQA, exceeding it by **nearly 9% relative** (0.613 vs. 0.563)—a result that decisively attributes the improvement to the learning algorithm rather than to model scale.

Our main contributions are as follows:

- **A unified and stable algorithm for agentic-search RL.** We propose DSPO, which resolves the twin problems of training instability and sample inefficiency that have long hindered RL for agentic search by integrating two mutually reinforcing principles within a single cohesive framework: **sequence-level optimization**, which delivers robust, low-variance policy updates, and **dynamic outcome-based filtering**, which guarantees a dense, consistently informative learning signal. The two are complementary by design—one governing gradient variance, the other signal density—and our ablations establish that neither alone suffices.
- **State-of-the-art performance under a rigorous, controlled protocol.** Evaluated across seven QA benchmarks under a strictly controlled protocol that holds the reward definition, training data, and retrieval component fixed, our DSPO-trained 7B model delivers a **34.1% relative improvement** over a comparable 7B baseline and, more strikingly, matches or outperforms its 14B counterpart on complex multi-hop QA, achieving a **nearly 9% relative gain** on HotpotQA (0.613 vs. 0.563). Because scale, data, and retrieval are held constant, these gains derive unambiguously from a superior learning algorithm rather than from additional model capacity.
- **Demonstrated stability, scalability, and transfer under a deliberately minimal retrieval setup.** We furnish extensive empirical evidence that DSPO traces a smooth, monotonic learning trajectory precisely where token-level baselines collapse. Because these results are obtained with nothing more than a standard BM25 retriever, the observed gains are attributable to the robustness of the algorithm itself rather than to a stronger retrieval component; and, as our further analyses establish, they persist under scaling to 14B and transfer to single-turn mathematical reasoning across two model families.

2 RELATED WORK

2.1 RL FOR LLMs

The landscape of RL for LLMs has evolved rapidly, progressing from foundational Reinforcement Learning from Human Feedback (RLHF) methods built on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler, direct-optimization frameworks such as DPO (Rafailov et al., 2023). A decisive shift toward value-free optimization was marked by Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which dispenses with a learned value model by deriving the advantage signal directly from group statistics. GRPO’s token-level objective, however, is a well-documented source of training instability (Liu et al., 2025; Cui et al., 2025), a deficiency that has prompted a succession of targeted refinements: GSPO reformulates the objective at the sequence level to match the unit of reward (Zheng et al., 2025); DAPO counters inefficient learning under sparse rewards through a dynamic outcome-based sampling mechanism (Yu et al., 2025); and, in a related vein, GMPO stabilizes the token-level objective via a geometric-mean aggregation that suppresses the influence of outliers (Zhao et al., 2025). Each of these advances, however, addresses a single failure mode in isolation, and, as our experiments confirm, the corresponding algorithms consequently still succumb to training collapse or plateau at a suboptimal performance ceiling. DSPO is the first to unify sequence-level optimization and dynamic outcome-based filtering within a single algorithm, jointly resolving the instability and signal sparsity that together afflict the training of autonomous search agents rather than confronting either in isolation.

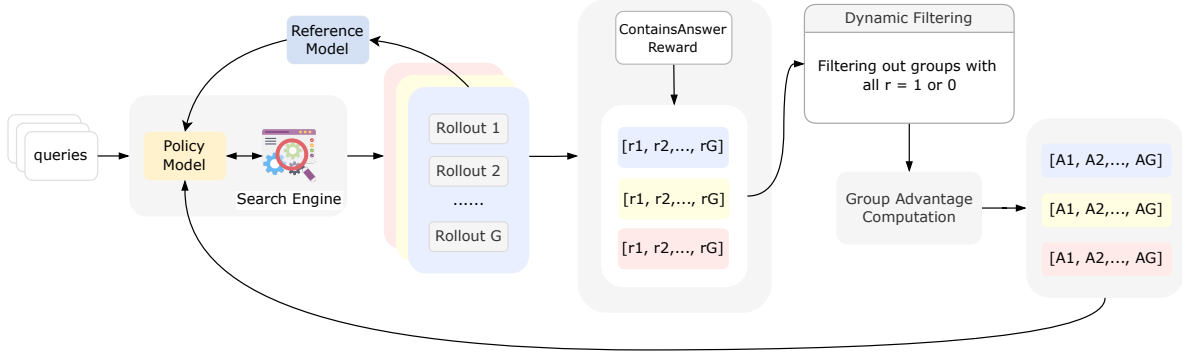


Figure 1: Overview of the DSPO training loop. For a given query, the policy model generates a group of G trajectories by interacting with the search environment, and each trajectory is assigned a sparse terminal reward. The **dynamic filter** then discards any group with homogeneous outcomes and continues sampling until the batch is filled, guaranteeing that every training batch carries an effective, non-degenerate advantage signal. The resulting advantages drive the policy update through the sequence-level objective, so that the two components—dense signal and stable optimization—are coupled within a single iteration rather than applied in isolation.

2.2 LLMs WITH AGENTIC RETRIEVAL

To mitigate the static-knowledge limitation of LLMs, Retrieval-Augmented Generation (RAG) couples the model with external retrievers that dynamically incorporate evolving information (Lewis et al., 2020; Gao et al., 2023). Classic RAG frameworks employ dense retrievers to fetch relevant documents and concatenate them into the LLM’s input for generation (Karpukhin et al., 2020); because such pipelines are fixed, however, they afford little autonomy in complex, multi-turn scenarios. Research has consequently converged on agentic paradigms in which LLMs act as autonomous agents that plan, search, and reason iteratively. ReAct interleaves reasoning and acting to let LLMs operate tools for tasks such as web navigation (Yao et al., 2023), while multi-agent systems such as AutoGen orchestrate collaborative workflows (Wu et al., 2024). A more recent and increasingly influential thrust foregrounds the integration of agentic RAG with RL, in which agents strengthen retrieval through learned decision-making: Wu et al. (2025) introduce Agentic Reasoning, which integrates external tools for streamlined LLM reasoning, while a family of RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) trains LLMs to interleave reasoning and search through RL alone. This end-to-end paradigm internalizes agentic capability within a single policy and sidesteps the engineering overhead of multi-agent orchestration, yet it remains persistently constrained by training instability and performance ceilings in the open-ended search domain. Our work confronts precisely these bottlenecks: DSPO provides a robust and efficient training framework that guarantees stable policy optimization, thereby enabling LLMs to acquire effective multi-turn search strategies with no recourse whatsoever to supervised demonstrations.

3 METHODOLOGY

This section develops DSPO from first principles. We first formalize agentic search as an RL problem and review the prior policy optimization algorithms, making explicit the structural limitations that render them inadequate in this setting. We then introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, detailing the two components that respectively secure training stability and training efficiency and establishing why each is necessary rather than merely helpful. Finally, we assemble these components into a single integrated training algorithm.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL-based training of LLMs commonly relies on policy gradient methods, of which PPO (Schulman et al., 2017) is the canonical instance for alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective evaluated on samples drawn from a behavior policy $\pi_{\theta_{\text{old}}}$. Averaged over tokens, this objective takes the form

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. A central drawback of PPO is its reliance on a separately trained value model to estimate token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015): this auxiliary model not only incurs substantial memory overhead but also constitutes an additional source of instability.

GRPO (Shao et al., 2024) was proposed precisely to remove this dependency. Rather than learning a value model, GRPO samples a group of G responses $\{y_i\}_{i=1}^G$ for each prompt x and estimates the advantage of each response by normalizing its reward against the group statistics. As in PPO, the resulting objective is optimized at the token level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage assigned to every token $y_{i,t}$ within a response y_i is set to the common sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

This construction lies at the crux of the mismatch our method later resolves: although the reward R_i is defined over the entire sequence, GRPO propagates the shared advantage $\hat{A}_i$ through a token-level importance ratio, thereby coupling an intrinsically sequence-level signal to token-level updates—a discrepancy whose destabilizing consequences we make precise below.

Agentic Search as a Markov Decision Process. We formalize the iterative process of agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP)—a formulation that renders the sequential structure of the task, and with it the precise locus of the credit-assignment problem, fully explicit.

- **State (s_t):** At turn t , the state s_t encodes the complete interaction history: the initial question q , every previously generated thought and search query, and all retrieved evidence returned by the environment. Because each retrieval must be interpreted and integrated into subsequent decisions, the state grows monotonically along the trajectory, inducing precisely the long-horizon dependencies that conventional token-level RL is ill-equipped to optimize.
- **Action (a_t):** An action is a complete textual segment generated by the policy π_θ , comprising free-form reasoning followed by a decision. It terminates either with a `</tool_call>` token, which issues a search and returns the results delimited by `</tool_response>`, or with an `</answer>` token, which terminates the trajectory. This structured action space compels the agent to learn not only *what* to generate but also *when* to search—a coupling that constitutes a principal source of the substantial variability in trajectory length.
- **Policy (π_θ):** The policy π_θ is the underlying LLM, which generates tokens autoregressively conditioned on the state. Although it conditions on the full history, the optimization target is computed exclusively over the model-generated thoughts and actions, with the environment-returned content masked out (Jin et al., 2025b). Optimizing π_θ therefore requires credit assignment across long sequences in which many intermediate reasoning steps receive no direct supervision or reward—a regime that renders stable sequence-level optimization indispensable rather than merely advantageous.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records the full sequence of reasoning and tool interactions for a given question, encompassing both model-generated actions and environment-returned search results. Because the reward is assigned to the trajectory as a whole, the optimization problem is intrinsically sequence-level: any early decision can propagate forward to determine the eventual answer’s correctness.
- **Reward ($R(\tau)$):** We adopt a sparse terminal reward that grants $R = 1$ when the final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we treat the search engine $\mathcal{S}$ as an explicit component of the environment, with which the policy LLM π_θ interacts by interleaving reasoning and retrieval calls. The overall objective is to obtain a policy that maximizes the expected terminal reward, regularized by a KL-divergence term that penalizes large deviations from a fixed reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x; \mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x; \mathcal{S}) || \pi_{\text{ref}}(y|x; \mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x; \mathcal{S})$ signifies that the trajectory y arises from a multi-step process in which the policy’s token generation is tightly coupled, at each retrieval step, with the evidence returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) have successfully cast agentic search as an RL problem, applying conventional algorithms such as PPO or GRPO within it encounters two structural obstacles, both sharply amplified by the open-ended nature of the search environment. The first is a fundamental mismatch between the unit at which reward is assigned (the sequence) and the unit at which optimization is performed (the token) (Zheng et al., 2025): over long trajectories, this discrepancy produces high-variance gradient estimates that accumulate turn after turn and frequently culminate in outright policy collapse. The second stems from the sparse binary reward: because many training batches contain exclusively successful or exclusively unsuccessful trajectories, the group-normalized advantage degenerates and conveys no learning signal whatsoever, drastically curtailing sample efficiency (Yu et al., 2025; Liu et al., 2025). These are not independent nuisances but the two dominant barriers to stable, efficient agentic-search RL, and DSPO is engineered to neutralize them in concert—the first through sequence-level optimization, the second through dynamic outcome-based filtering.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO advances two tightly coordinated innovations over prior methods: (1) it performs policy optimization at the sequence level, aligning the training objective with the trajectory-based reward structure and thereby eliminating the reward–optimization mismatch outright; and (2) it introduces a dynamic filtering mechanism that guarantees every training batch carries a high-quality, non-degenerate learning signal. These innovations target orthogonal failure modes—the first gradient variance, the second signal density—and their integration, depicted in Figure 1, constitutes the complete training procedure.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following the sequence-level formulation of GSPO (Zheng et al., 2025), we replace the unstable token-level importance ratio with a length-normalized sequence-level counterpart. The sequence-level importance ratio $s_i(\theta)$ for a response y_i is defined as the geometric mean of its constituent token-level ratios, which equivalently corresponds to the length-normalized exponential of their summed log-ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\text{old}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\text{old}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length normalization is essential rather than cosmetic: through the exponent $1/|y_i|$, it confines $s_i(\theta)$ to a consistent numerical range irrespective of sequence length, suppressing the variance that would otherwise accumulate across long trajectories and rendering the subsequent clipping operation both stable and well-calibrated.

Gradient Analysis. A direct comparison of gradients makes the source of DSPO’s stability precise. The (unclipped) gradient of the token-level GRPO objective scales each token’s log-probability gradient by a noisy, token-specific weight $r_{i,t}(\theta)$, so that estimation error compounds multiplicatively across the many tokens of a long trajectory. Our sequence-level objective instead scales the aggregate log-probability gradient of the entire sequence by a single, far more stable sequence-level weight $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying a single holistic correction factor to the entire trajectory, DSPO eliminates the token-level noise accumulation that destabilizes prior methods and thereby yields fundamentally more stable training. This stabilization is complemented by the dynamic filtering mechanism, which guarantees a well-defined advantage signal $\hat{A}_i$ by admitting only those rollout groups that contain both successes and failures, thereby purging the wasted, signal-free samples that pervade sparse-reward environments. The two mechanisms consequently act on orthogonal axes—one controlling gradient variance, the other controlling signal density—and it is their combination, not either component alone, that makes stable, high-performance training attainable.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary structure of our reward function poses an acute challenge for group-based advantage estimation. Whenever all G responses in a group share the same outcome—uniformly correct ($R = 1$) or uniformly incorrect ($R = 0$)—the group’s reward variance vanishes and the normalized advantage $\hat{A}_i$ collapses to zero or becomes undefined. Such degenerate groups contribute no useful gradient and squander the entire computation expended in generating them, a cost that becomes prohibitive under the sparse rewards characteristic of search.

DSPO overcomes this pathology through a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, we retain only those trajectory groups that exhibit a mixture of successful and unsuccessful outcomes; formally, a group $\{y_i\}_{i=1}^G$ is admitted for training if and only if its rewards $\{R_i\}_{i=1}^G$ satisfy

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition guarantees that the reward variance within every training group is strictly positive, so that the group-normalized advantage is invariably well-defined and informative. In effect, the filter actively curates a high-quality batch for each policy update, transforming an otherwise sparse reward landscape into a dense and efficient learning signal—all without altering the underlying reward definition.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Integrating the two components yields the complete DSPO objective. For each admitted group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ by group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant added for numerical stability. The policy π_θ is then updated by maximizing

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined above as the geometric mean of the token-level ratios. Crucially, during likelihood computation we apply loss masking to every token retrieved from the search tool, following Jin et al. (2025b); this restriction confines the gradient to model-generated content and ensures that the policy learns to *reason over* external knowledge rather than merely to reproduce it. The complete procedure, which interleaves filtered sampling with sequence-level updates within each training step, is specified in Algorithm 1.

4 EXPERIMENTS

We now present a comprehensive empirical evaluation designed to substantiate the effectiveness of Dynamic-filter Sequence-level Policy Optimization (DSPO). The experiments are organized around three claims, which the results collectively confirm: that DSPO (1) attains state-of-the-art performance across a suite of challenging question-answering benchmarks; (2) exhibits markedly superior training stability, wholly averting the catastrophic collapse that afflicts baseline methods; and (3) derives its gains from the synergy of its two components, each of which our controlled ablations reveal to be indispensable rather than merely incremental.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we adopt the prompt template shown in Table 1, which organizes the model’s actions during search into three interleaved components—reasoning delimited by `<think>`, retrieval delimited by

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
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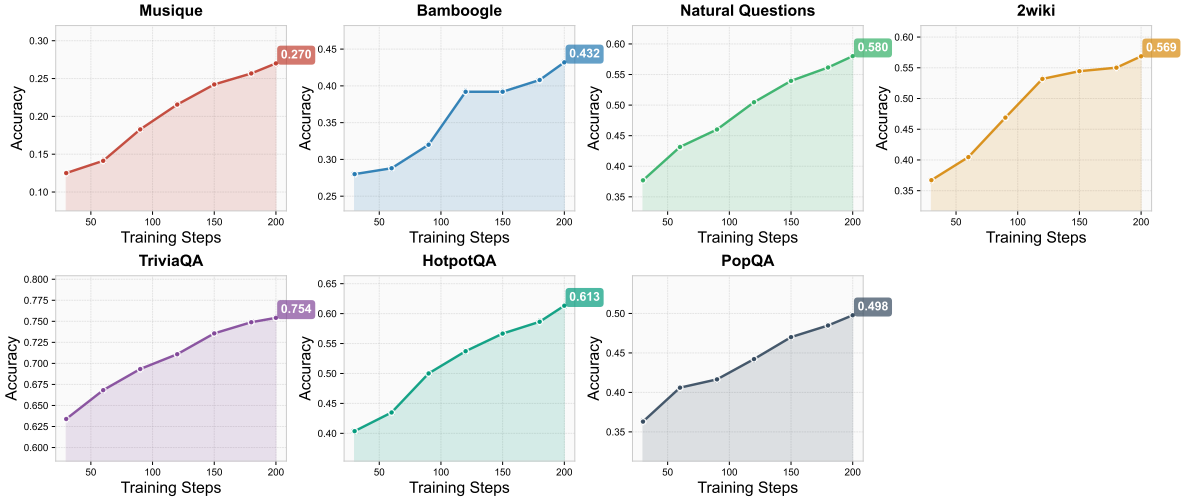


Figure 2: **Validation performance of DSPO across seven benchmarks during training.** Held-out accuracy rises steadily and monotonically on every one of the seven benchmarks, confirming that DSPO’s gains in training reward translate directly into stronger generalization rather than into reward hacking, and that the method converges to a robust, transferable search-and-reasoning policy.

<tool_call>, and the final response delimited by <answer>—thereby imposing an explicit, parseable structure on every trajectory.

Benchmarks and Baselines. To ensure a rigorous and directly comparable evaluation, our experimental design adheres strictly to the established protocol of Search-R1 (Jin et al., 2025b). We train on a composite dataset comprising the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), and we assess generalization on the test sets of seven diverse QA benchmarks that span both single-hop and multi-hop reasoning: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022). This breadth allows us to characterize DSPO’s behavior across the full difficulty spectrum of open-domain question answering, from single-fact lookup to compositional multi-hop reasoning.

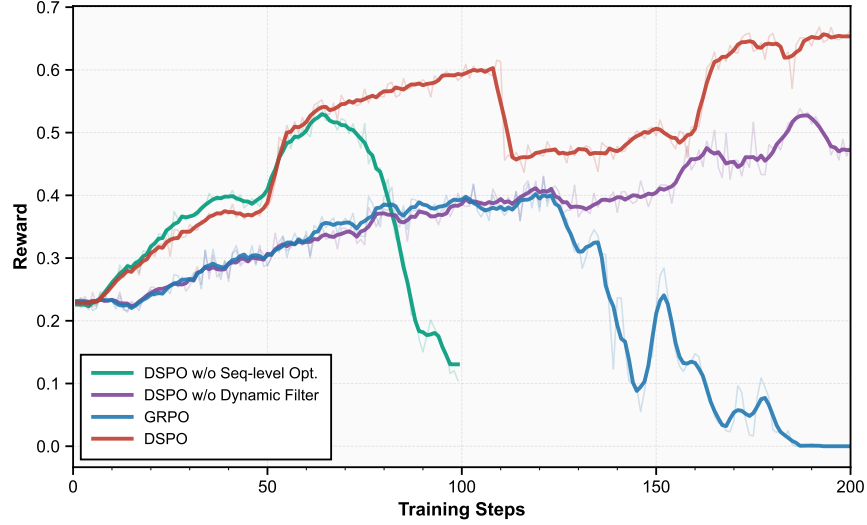


Figure 3: **Training reward dynamics of DSPO and its ablations.** The comparative learning curves reveal a sharp, design-dictated separation. DSPO (red) improves stably and monotonically to the highest reward; the token-level variants (green, blue) undergo catastrophic policy collapse after only a brief initial rise; and the sequence-level variant without the dynamic filter (purple) remains stable yet plateaus at a markedly suboptimal level. This three-way dissociation cleanly isolates the distinct role of each component: sequence-level updates govern whether training remains stable, whereas the dynamic filter governs how high performance ultimately climbs.

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: The prompt template employed throughout our experiments, which directs the model to interleave reasoning, retrieval, and answering within the designated structural tags.

Our comparison suite deliberately pairs strong external baselines with controlled internal ablations, so as to isolate both the overall advantage of DSPO and the individual contribution of each of its components. The external baselines are the Qwen2.5-7B and 14B models trained with PPO and GRPO under the Search-R1 framework (Jin et al., 2025b;a). To decompose our method, we further evaluate two ablations, each disabling exactly one component: (1) **DSPO w/o dynamic filter**, which reduces to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to the strong token-level policy DAPO (Yu et al., 2025). This construction lets us attribute any observed performance difference to a single, identifiable mechanism.

For implementation, every training run initializes from the Qwen2.5-7B-Instruct checkpoint. Our experiments are built on the VeRL framework (Sheng et al., 2025), adapting its provided Search-R1-like example code and scripts to our methodology. Because our setup employs a modified reward function, we deliberately refrain from citing the published Search-R1 numbers directly; instead, we retrain the PPO and GRPO baselines (Jin et al., 2025b;a) under our exact experimental conditions, and these retrained models constitute our primary external benchmarks. This protocol guarantees that every reported comparison holds the reward definition, training data, and retrieval setup fixed, so that any observed difference in performance is attributable to the learning algorithm alone.

Implementation and Evaluation. To cleanly separate the contribution of the learning algorithm from that of retrieval quality, all RL experiments deliberately employ a standard, unlearned BM25 retriever. This controlled choice ensures that the observed improvements are attributable to the learned policy rather than to a stronger retrieval component, so that the gains reported here reflect the algorithm operating under a deliberately minimal retrieval setup. Across all methods, models are trained with a sparse binary reward defined by substring Exact Match (subEM) of the final

answer, and subEM likewise serves as the primary evaluation metric, keeping the training signal and the evaluation criterion consistent throughout.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 presents a comprehensive comparison that consolidates our main results and ablation study within a single view. Because DSPO’s two components act synergistically, juxtaposing the full method against both external state-of-the-art baselines and its own ablated variants in one table is the most direct means of establishing two distinct conclusions at once: the overall superiority of DSPO relative to prior work, and the indispensability of each of its constituent components.

Table 2: Comprehensive comparison of DSPO against external baselines and ablation variants across seven QA benchmarks, reported as subEM. The external baselines are the Search-R1 models (7B and 14B) trained with GRPO and PPO (Jin et al., 2025b;a); the two ablations each disable a single component (‘w/o dynamic filter’ and ‘w/o seq-level opt.’). To guarantee a consistent comparison, all baselines were retrained and re-evaluated under our adjusted reward, with the originally published Search-R1 EM scores shown in parentheses for reference only. Best results are in **bold** and second-best are underlined. The DSPO-trained 7B model attains the best or second-best score on every one of the seven benchmarks and the highest average overall, matching the far larger 14B baselines despite using half the parameters.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results in Table 2 establish DSPO’s clear and consistent superiority over comparable methods. Our DSPO-trained 7B agent attains an average subEM of 0.531—the highest among all evaluated models and a 34.1% relative improvement over the same-sized Search-R1 (GRPO, 7B) baseline at 0.377. More striking still, this 7B agent matches or slightly exceeds the average of the far larger Search-R1 14B models under both GRPO (0.530) and PPO, and on the demanding multi-hop HotpotQA benchmark it reaches 0.613 against the 14B GRPO model’s 0.563—a nearly 9% relative gain. That a 7B policy can equal or surpass a 14B one under an identical retrieval setup is compelling evidence that DSPO’s gains stem from a more effective and stable learning algorithm rather than from any reliance on model scale.

Analysis of Ablations. The ablation results in Table 2 demonstrate unambiguously that both of DSPO’s components are indispensable, each accounting for a large, well-localized share of the full method’s performance. Removing the dynamic filter (‘w/o dynamic filter’) proves the more damaging of the two: the average subEM collapses from 0.531 to 0.313. Deprived of the filter, the sequence-level objective is repeatedly presented with homogeneous-reward batches and starved of any usable learning signal—a failure that directly confirms the filter’s role as the mechanism sustaining an informative advantage signal. Ablating sequence-level optimization (‘w/o sequence-level opt.’) is likewise markedly detrimental, reducing the average to 0.406. Although this token-level variant outperforms the filter-less one, it falls well short of the full model and, as the next section shows, is additionally prone to catastrophic training instability. Taken together, these two outcomes establish that the components are complementary rather than redundant: sequence-level updates supply the stability, the dynamic filter supplies the dense signal, and only their combination realizes the full 0.531 average.

Beyond these aggregate metrics, DSPO gives rise to sophisticated search behaviors—recognizing irrelevant results, reformulating queries, and verifying answers across multiple turns (see Appendix A for detailed trajectory examples).

Notably, every one of these behaviors emerges from pure RL training under DSPO, without any supervised demonstration of search strategy whatsoever.

4.3 ANALYSIS OF TRAINING DYNAMICS

To substantiate our claims regarding stability and efficiency, we examine the training reward dynamics of DSPO alongside its ablations and key baselines. Figure 3 visualizes these trajectories, and the pronounced separation among them directly corroborates our two core design choices.

DSPO (red) ascends smoothly and monotonically, converging efficiently to the highest reward level and thereby confirming the stability conferred by the sequence-level objective. The **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—stand in stark contrast: after only a brief initial rise, their rewards collapse catastrophically, a direct manifestation of the high-variance, token-level gradient updates identified in our earlier gradient analysis. **DSPO w/o Dynamic Filter** (purple), which retains sequence-level updates but forgoes the dense learning signal, remains stable yet plateaus at a substantially suboptimal ceiling. These three regimes thus dissociate the two components cleanly—sequence-level updates govern whether training remains stable, while the dynamic filter governs how high performance can ultimately climb—and only their conjunction secures both stability and a high asymptote.

To verify that these reward improvements reflect genuine generalization rather than reward hacking, we track validation performance on the key benchmarks throughout training. As Figure 2 shows, DSPO’s validation accuracy on NQ, HotpotQA, and the remaining benchmarks rises consistently and in lockstep with its reward curve. This tight correspondence between training reward and held-out accuracy confirms that the agent acquires a genuinely generalizable search-and-reasoning policy rather than exploiting the reward signal.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To probe the robustness of our approach beyond the primary 7B setting, we extend the evaluation along two complementary axes: scaling to a larger model, and transfer to an entirely different reasoning domain.

Scalability to Larger Models. We first ask whether DSPO’s stability benefits persist at larger parameter scales, training Qwen2.5-14B-Instruct under an identical protocol. As detailed in Table 3, DSPO scales cleanly and without exception: the DSPO-trained 14B model attains an average accuracy of 60.6%, outperforming the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%** and improving over that baseline on all seven benchmarks, with per-benchmark relative gains ranging from +5.5% to +38.8%. The advantage is systematic rather than isolated, confirming that DSPO effectively converts additional model capacity into higher accuracy and that its algorithmic benefits compound with—rather than wash out at—larger scale.

Table 3: Scalability analysis on Qwen2.5-14B-Instruct. The DSPO-trained 14B model surpasses the GRPO-14B baseline on every one of the seven benchmarks, with per-benchmark relative gains ranging from +5.5% to +38.8%; the final column reports the relative gain of DSPO (14B) over GRPO (14B), and the DSPO (7B) column is included to situate the scaling effect against the primary 7B model. Best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. We next assess whether DSPO’s benefits extend beyond multi-turn search to the distinct setting of single-turn mathematical reasoning, evaluating models drawn from both the Qwen2.5 and Qwen3 families. As reported in Table 4, DSPO surpasses GRPO in every configuration tested—on Math500 with Qwen2.5-Math-7B (0.798 vs. 0.772) and on Olympiad-Bench with Qwen3-4B (0.755 vs. 0.728)—improving

consistently across both model families and sizes. Because the sparse-reward and unstable-gradient pathologies that DSPO addresses are in no way specific to search, these gains indicate that its underlying principles transfer to general reasoning domains rather than remaining confined to the agentic-retrieval setting.

Table 4: Generalization to single-turn mathematical reasoning across the Qwen2.5 and Qwen3 families. DSPO outperforms GRPO in both configurations, spanning two model families and sizes and evaluated at the indicated number of training steps; the final column reports the relative gain. Best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

In this work, we resolved the twin problems of training instability and sample inefficiency that have long hindered RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which secures robust training through two mutually reinforcing components: sequence-level optimization that forestalls catastrophic policy collapse, and a dynamic outcome-based filter that converts sparse rewards into a consistently effective learning signal. Across seven QA benchmarks, DSPO not only delivers state-of-the-art performance—improving over a comparable 7B baseline by 34.1% relative and matching or exceeding the far larger 14B baselines—but also exhibits markedly superior training stability; and, as our further analyses establish, these advantages persist under scaling to 14B and transfer to single-turn mathematical reasoning.

By enabling robust training from environmental feedback alone, DSPO furnishes a practical and efficient blueprint for building capable LLM agents without recourse to costly expert demonstrations. This stable foundation opens clear avenues for future work, including the integration of stronger, learned retrievers and the extension of DSPO to complex, multi-tool settings. Moreover, because sparse rewards and unstable policy gradients are by no means unique to search—as our mathematical-reasoning experiments already begin to demonstrate—we anticipate that DSPO’s principles will yield comparable gains in domains such as mathematics and code generation, a direction we regard as especially promising. We contend that the two core tenets of DSPO—matching the unit of optimization to the unit of reward, and guaranteeing the density of the learning signal—constitute durable design principles for the next generation of autonomous AI agents.

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A APPENDIX

To illuminate the search strategies acquired through training, Table A.1 presents representative trajectories produced by our model. These examples concretely instantiate three distinct learned competencies: (1) reformulating queries when an initial search returns irrelevant results, (2) verifying information through follow-up searches, and (3) extracting comprehensive answers from retrieved evidence—each of which emerges from RL training alone, without any supervised demonstration of the underlying strategy.

Table A.1: Representative trajectories illustrating the search-and-reasoning capabilities our model acquires through DSPO training. The model exhibits sophisticated behaviors, including (a) recognition of irrelevant results, (b) query reformulation, and (c) verification through follow-up search—all acquired via pure RL, with no supervised demonstrations whatsoever.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin's occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American journalist, attorney, soldier, and politician. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician